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Database Solution For SwiftShip Deliveries

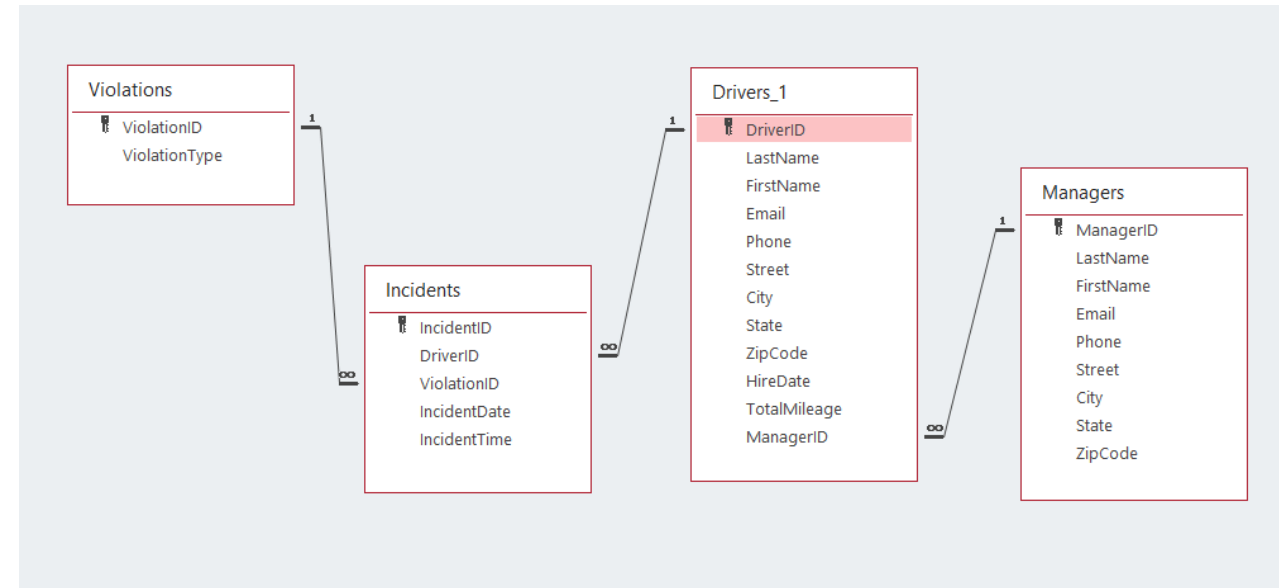
Presentation by Carter Consulting Company

Project Overview

- SwiftShip Deliveries aims to provide fast and reliable shipment services for businesses and individuals.
- The Microsoft Access database we've developed tracks the safety incidents of SwiftShip's delivery drivers in an effort to help enhance driver safety.
- This database will help improve SwiftShip's operations by making data management and analysis of driver behavior much easier to handle.

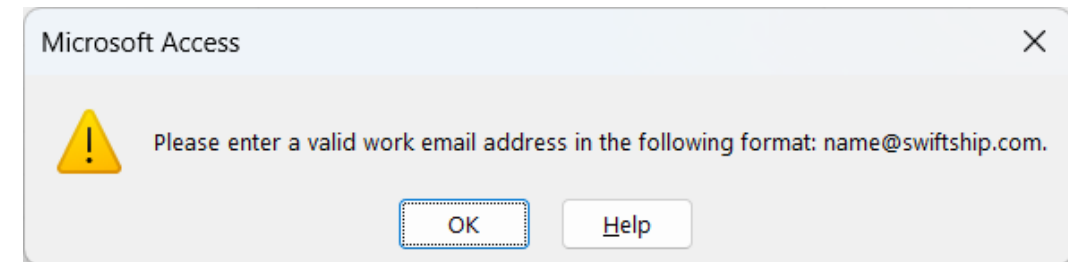
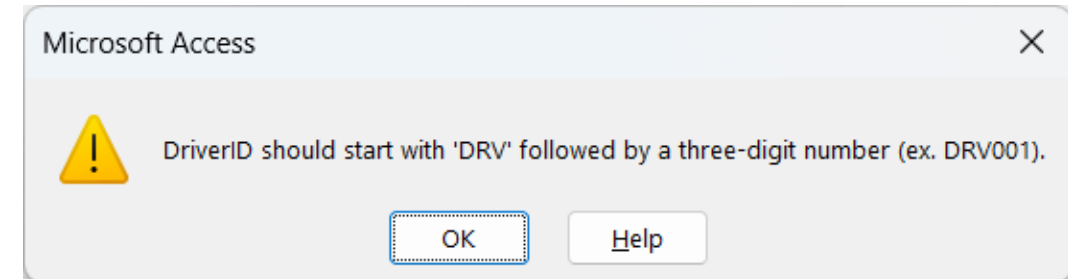
Database Structure

- The database contains 4 tables:
 - Drivers
 - Managers
 - Incidents
 - Violations
- The database utilizes 3 one-to-many relationships:
 - Each manager can supervise multiple drivers.
 - Each driver can be associated with multiple incidents.
 - Each violation type can be referenced on multiple incidents.



Data Accuracy & Integrity

- Validation rules are used throughout the database to ensure the accuracy of the data being entered and to prevent invalid entries.
- Input masks were also utilized to make entering various information more straightforward.
- Some examples include:
 - Rule to ensure proper formatting of primary IDs for each table.
 - Rule to ensure that the email entered is a company email.
 - Input mask utilized for entering phone numbers.
 - Lookup dropdown utilized to display existing records (in this case manager records).



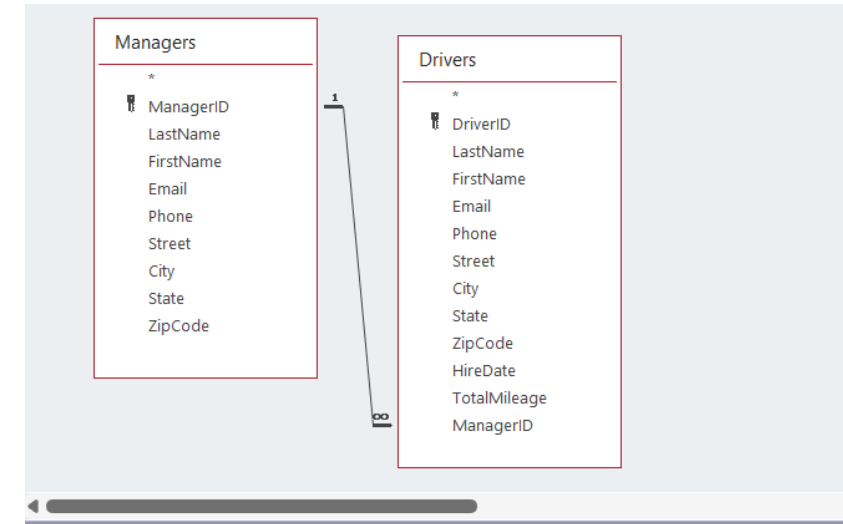
Phone

ManagerID

	▼
MGR01	
MGR02	
MGR03	

Querying Information

- The database makes use of queries to retrieve and calculate various information.
- Some queries gather information from one table while others gather from multiple tables.
- Some examples include:
 - A query that displays all managers and the number of drivers that they supervise.
 - A query that calculates the total number of incidents per violation type.
 - A query that lists drivers hired in the last 6 months.



Field:	ManagerID	LastName	FirstName	DriverID
Table:	Managers	Managers	Managers	Drivers
Total:	Group By	Group By	Group By	Count
Sort:	Ascending			
Show:	☒	☒	☒	☒
Criteria:				
or:				



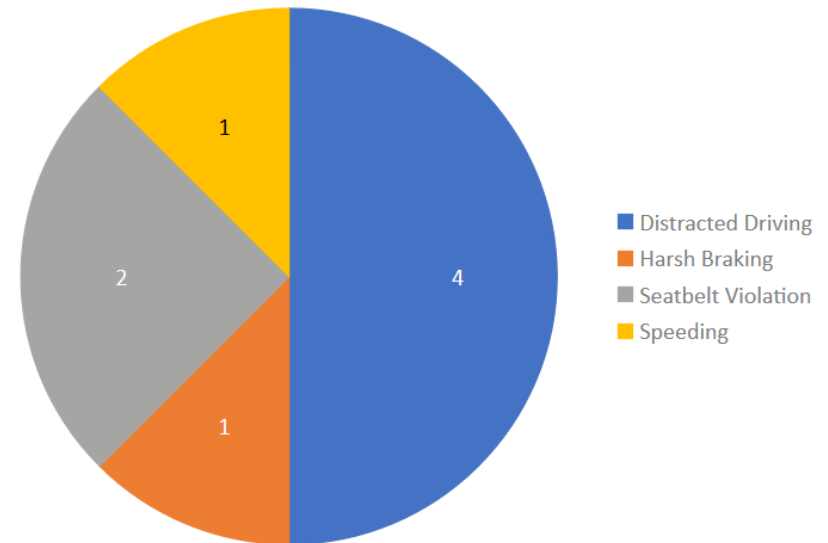
Managers and Number of Drivers Supervised			
ManagerID	LastName	FirstName	# of Drivers Supervised
MGR01	Barnes	Emily	2
MGR02	Clark	Daniel	2
MGR03	Allen	Jessica	1

Viewing Reports

- The database generates informative reports that present data in a way that is visually appealing and easy to analyze.
- One of the reports in the database generates a summary of incidents by violation type based on the total incidents per violation type query. It also creates a pie graph, which is helpful for visualizing the data.

Total Incidents Per Violation Type Report

Violation Type	# of Incidents
Distracted Driving	4
Seatbelt Violation	2
Speeding	1
Harsh Braking	1
Total	8



Navigation Interface

- It is important to be able to easily navigate through the various objects in the database.
- The database has a navigation form with a two-level tab layout. This can be used to navigate between all the different forms and reports.
- The database also utilizes macros, which are shortcuts used to perform repetitive actions or commands. These can be found on the navigation pane.
- Some examples include:
 - A macro to add a new record in a form.
 - A macro that closes all open objects.
 - Macros to open or export various objects in the database.

The screenshot displays the 'Navigation Form' interface. At the top, there is a header bar with a document icon and the title 'Navigation Form'. Below this, a tabbed interface is shown with two main tabs: 'Forms' and 'Reports'. Under the 'Forms' tab, there are three sub-tabs: 'Incident Form', 'Driver Form', and 'Manager Form'. The 'Incident Form' is currently selected and active. The form contains several input fields: 'IncidentID' (text box with 'INC0001'), 'DriverID' (dropdown menu with 'DRV001'), 'ViolationID' (dropdown menu with '01'), 'ViolationType' (text box with 'Speeding'), 'IncidentDate' (text box with '7/25/2023'), and 'IncidentTime' (text box with '9:30 AM'). To the right of the form, there is a 'Macros' section with a list of shortcuts, each preceded by a small icon. The macros include: 'Add New Driver', 'Add New Incident', 'Add New Manager', 'Close All and Return to Nav Form', 'Close All Open Objects', 'Close Database', 'Export Incident Records', 'Open All Forms', 'Open All Tables', 'Preview All Reports', 'View Contact Info', and 'View Violation Types'.

Navigation Form	
Forms Reports	
Incident Form Driver Form Manager Form	
Incident Form	
IncidentID	INC0001
DriverID	DRV001
ViolationID	01
ViolationType	Speeding
IncidentDate	7/25/2023
IncidentTime	9:30 AM

Macros

- Add New Driver
- Add New Incident
- Add New Manager
- Close All and Return to Nav Form
- Close All Open Objects
- Close Database
- Export Incident Records
- Open All Forms
- Open All Tables
- Preview All Reports
- View Contact Info
- View Violation Types

Conclusion

- Our database solution offers SwiftShip Deliveries a streamlined approach to managing driver safety incidents, helping to enhance overall operations.
- Features such as validation rules, queries, and informative reports help ensure data accuracy and ease of use within the database.
- We are confident that this solution will have a positive impact on SwiftShip's safety and efficiency standards.

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◦ **Thank You
For Your Time!**